

Functional Equations and Inequalities in Several Variables

Stefan Czerwik

Silesian University of Technology, Poland

Functional Equations and Inequalities in Several Variables



World Scientific

New Jersey • London • Singapore • Hong Kong

Published by

World Scientific Publishing Co. Pte. Ltd.

P O Box 128, Farrer Road, Singapore 912805

USA office: Suite 1B, 1060 Main Street, River Edge, NJ 07661

UK office: 57 Shelton Street, Covent Garden, London WC2H 9HE

British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library.

FUNCTIONAL EQUATIONS AND INEQUALITIES IN SEVERAL VARIABLES

Copyright © 2002 by World Scientific Publishing Co. Pte. Ltd.

All rights reserved. This book, or parts thereof, may not be reproduced in any form or by any means, electronic or mechanical, including photocopying, recording or any information storage and retrieval system now known or to be invented, without written permission from the Publisher.

For photocopying of material in this volume, please pay a copying fee through the Copyright Clearance Center, Inc., 222 Rosewood Drive, Danvers, MA 01923, USA. In this case permission to photocopy is not required from the publisher.

ISBN 981-02-4837-7

Printed in Singapore by World Scientific Printers (S) Pte Ltd

Preface

Functional equations have substantially grown to become an important branch of mathematics, particularly during the last two decades, with its special methods, a number of interesting results and several applications.

Many aspects of functional equations containing several variables may be found by the reader in the books of J. Aczél and J. Dhombres [1], M. Kuczma [123] and D. H. Hyers, G. Isac and Th. M. Rassias [97]. For more information about the history of functional equations the reader may consult Aczél [2] and Dhombres [43].

This book combines the classical theory and examples as well as recentmost results in the subject.

The recent book consists of three parts. The first one is devoted to additive functions and convex functions defined on linear spaces endowed with so-called semilinear topologies. Basic results concerning important functional equations are also included in the first part. In the second part of this book we study the problem of stability of functional equations of several variables is considered. This problem has originally been posed by S. Ulam in 1940. In 1941 D. H. Hyers gave a significant partial solution to this problem in his paper [100]. In 1978, Th. M. Rassias [178] generalized the Hyers' result, a fact which rekindled interest in the field. Since then a number of articles have appeared in the literature. Such type of stability is now called the Ulam-Hyers-Rassias stability of functional equations.

Finally, functional equations with set-valued functions are dealt

with in Part 3.

A systematic exposition of functional equations with set-valued functions is still lacking in the mathematical literature, as the result are scattered over several articles. An attempt is made here to fill this gap.

As for the background, we assume that the reader has some knowledge of the mathematical analysis, algebra, set theory and topology.

The specialized notation we use is the same as in the relevant articles. Standard symbols $\mathbb{Z}_+$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, ■ denote the sets of positive integers, integers, rationals, reals, complex numbers and the end of the proof, respectively.

Many other symbols are introduced in the text of the book.

The list of references at the end of the book is by no means complete. We have included only those articles and books that we referred to in the text of the book.

It is a pleasant duty to express here our gratitude to Professor Themistocles M. Rassias, for fruitful suggestions and comments.

We wish to express our thanks to World Scientific Publishing Co. for an efficient and agreeable cooperation.

Stefan Czerwik
June, 2001

Contents

Preface	v
 PART I Functional Equations and Inequalities in Linear Spaces	 1
1 Linear spaces and semilinear topology	3
2 Convex functions	9
3 Lower hull of a convex function	19
4 Theorems of Bernstein-Doetsch, Piccard and Mehdi	23
5 Some set classes of continuous and J-convex functions	29
6 Cauchy's exponential equation	35
7 D'Alembert's equation on abelian groups	43
8 D'Alembert's equation on topological groups	49
9 Polynomial functions and their extensions	65
10 Quadratic mappings	89

11 Quadratic equation on an interval	113
12 Functional equations for quadratic differences	121
PART II Ulam-Hyers-Rassias Stability of Functional Equations	127
13 Additive Cauchy equation	129
14 Multiplicative Cauchy equation	141
15 Jensen's functional equation	147
16 Pexider's functional equation	153
17 Gamma functional equation	157
18 D'Alembert's and Lobachevski's functional equations	161
19 Stability of homogeneous mappings	169
20 Quadratic functional equation	185
21 Stability of functional equations in function spaces	203
22 Cauchy difference operator in L^p spaces	213
23 Pexider difference operator in L^p spaces	221
24 Cauchy and Pexider operators in X_λ spaces	229
25 Stability in the Lipschitz norms	235
26 Round-off stability of iterations	245

27 Quadratic difference operator in L^p spaces	251
 PART III Functional Equations in Set-Valued Functions	 261
28 Cauchy's set-valued functional equation	263
29 Jensen's functional equation	279
30 Pexider's functional equation	287
31 Quadratic set-valued functions	293
32 Subadditive set-valued functions	301
33 Superadditive set-valued functions and generalization of Banach-Steinhaus theorem	323
34 Hahn-Banach type theorem and applications	331
35 Subquadratic set-valued functions	345
36 K-convex and K-concave set-valued functions	355
37 Iteration semigroups of set-valued functions	379
References	387
Index	407